Advanced Topics in ACP

https://taroccoesbrocco.github.io/ACP2026.html

Solutions to selected exercises from Lecture 1 on the λ-calculus

Giulio Guerrieri

Institut für Informatik, Universität Innsbruck. giulio.guerrieri@uibk.ac.at

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Non-assessed exercises from Lecture 1

(https://taroccoesbrocco.github.io/ACP2026/Lecture01.pdf)

Exercise 1

Prove the following facts, using ND and $\mathsf{ND}_{\mathrm{seq}}$.

1. $\vdash X \Rightarrow ((X \Rightarrow Y) \Rightarrow Y)$.
2. $(X \Rightarrow Y) \Rightarrow (X \Rightarrow Z) \vdash Y \Rightarrow X \Rightarrow Z$.
3. $(X \Rightarrow Y) \Rightarrow X \vdash Y \Rightarrow X$.
4. $X \Rightarrow (Y \Rightarrow Z) \vdash Y \Rightarrow X \Rightarrow Z$.
5. $X \Rightarrow Y \Rightarrow Z,\ X \Rightarrow Y \vdash X \Rightarrow Z$.
6. $(X \Rightarrow X) \Rightarrow Y \vdash (Y \Rightarrow Z) \Rightarrow Z$.

Solution to Exercise 1

1. In ND and $\mathsf{ND}_{\mathrm{seq}}$, respectively:

$$\dfrac{\dfrac{\dfrac{[X \Rightarrow Y]^{\circ} \qquad [X]^{*}}{Y}\Rightarrow_e}{(X \Rightarrow Y) \Rightarrow Y}\Rightarrow_i^{\circ}}{X \Rightarrow ((X \Rightarrow Y) \Rightarrow Y)}\Rightarrow_i^{*} \qquad \dfrac{\dfrac{\dfrac{\dfrac{}{X, X \Rightarrow Y \vdash X \Rightarrow Y}\,\mathsf{ax} \qquad \dfrac{}{X, X \Rightarrow Y \vdash X}\,\mathsf{ax}}{X, X \Rightarrow Y \vdash Y}\Rightarrow_e}{X \vdash (X \Rightarrow Y) \Rightarrow Y}\Rightarrow_i}{\vdash X \Rightarrow ((X \Rightarrow Y) \Rightarrow Y)}\Rightarrow_i$$

2. In ND:

$$\dfrac{\dfrac{(X \Rightarrow Y) \Rightarrow (X \Rightarrow Z) \qquad \dfrac{[Y]^{*}}{X \Rightarrow Y}\Rightarrow_i}{X \Rightarrow Z}\Rightarrow_e}{Y \Rightarrow X \Rightarrow Z}\Rightarrow_i^{*}$$

In $\mathsf{ND}_{\mathrm{seq}}$:

$$\dfrac{\dfrac{\dfrac{}{(X \Rightarrow Y) \Rightarrow (X \Rightarrow Z), Y \vdash (X \Rightarrow Y) \Rightarrow (X \Rightarrow Z)}\,\mathsf{ax} \qquad \dfrac{\dfrac{}{(X \Rightarrow Y) \Rightarrow (X \Rightarrow Z), X, Y \vdash Y}\,\mathsf{ax}}{(X \Rightarrow Y) \Rightarrow (X \Rightarrow Z), Y \vdash X \Rightarrow Y}\Rightarrow_i}{(X \Rightarrow Y) \Rightarrow (X \Rightarrow Z), Y \vdash X \Rightarrow Z}\Rightarrow_e}{(X \Rightarrow Y) \Rightarrow (X \Rightarrow Z) \vdash Y \Rightarrow X \Rightarrow Z}\Rightarrow_i^{*}$$

3. In ND and $\mathsf{ND}_{\mathrm{seq}}$, respectively:

$$\dfrac{\dfrac{(X \Rightarrow Y) \Rightarrow X \qquad \dfrac{[Y]^*}{X \Rightarrow Y}\Rightarrow_i}{X}\Rightarrow_e}{Y \Rightarrow X}\Rightarrow_i^* \qquad \dfrac{\dfrac{\dfrac{}{(X \Rightarrow Y) \Rightarrow X, Y \vdash (X \Rightarrow Y) \Rightarrow X}\,\mathsf{ax} \qquad \dfrac{\dfrac{}{(X \Rightarrow Y) \Rightarrow X, X, Y \vdash Y}\,\mathsf{ax}}{(X \Rightarrow Y) \Rightarrow X, Y \vdash X \Rightarrow Y}\Rightarrow_i}{(X \Rightarrow Y) \Rightarrow X, Y \vdash X}\Rightarrow_e}{(X \Rightarrow Y) \Rightarrow X \vdash Y \Rightarrow X}\Rightarrow_i$$

4. In ND:

$$\dfrac{\dfrac{\dfrac{X \Rightarrow (Y \Rightarrow Z) \qquad [X]^\circ}{Y \Rightarrow Z}\Rightarrow_e \qquad [Y]^*}{Z}\Rightarrow_e}{X \Rightarrow Z}\Rightarrow_i^\circ}{Y \Rightarrow X \Rightarrow Z}\Rightarrow_i^*$$

In $\mathsf{ND}_{\mathrm{seq}}$:

$$\dfrac{\dfrac{\dfrac{\dfrac{}{X \Rightarrow (Y \Rightarrow Z), X, Y \vdash X \Rightarrow (Y \Rightarrow Z)}\,\mathsf{ax} \qquad \dfrac{}{X \Rightarrow (Y \Rightarrow Z), X, Y \vdash X}\,\mathsf{ax}}{X \Rightarrow (Y \Rightarrow Z), X, Y \vdash Y \Rightarrow Z}\Rightarrow_e \qquad \dfrac{}{X \Rightarrow (Y \Rightarrow Z), X, Y \vdash Y}\,\mathsf{ax}}{X \Rightarrow (Y \Rightarrow Z), X, Y \vdash Z}\Rightarrow_e}{X \Rightarrow (Y \Rightarrow Z), Y \vdash X \Rightarrow Z}\Rightarrow_i}{X \Rightarrow (Y \Rightarrow Z) \vdash Y \Rightarrow X \Rightarrow Z}\Rightarrow_i$$

5. In ND:

$$\dfrac{\dfrac{\dfrac{X \Rightarrow Y \Rightarrow Z \qquad [X]^*}{Y \Rightarrow Z}\Rightarrow_e \qquad \dfrac{X \Rightarrow Y \qquad [X]^*}{Y}\Rightarrow_e}{Z}\Rightarrow_e}{X \Rightarrow Z}\Rightarrow_i^*$$

In $\mathsf{ND}_{\mathrm{seq}}$:

$$\dfrac{\dfrac{\dfrac{\dfrac{}{X \Rightarrow Y \Rightarrow Z, X \Rightarrow Y, X \vdash X \Rightarrow Y \Rightarrow Z}\,\mathsf{ax} \qquad \dfrac{}{X \Rightarrow Y \Rightarrow Z, X \Rightarrow Y, X \vdash X}\,\mathsf{ax}}{X \Rightarrow Y \Rightarrow Z, X \Rightarrow Y, X \vdash Y \Rightarrow Z}\Rightarrow_e \qquad \dfrac{\dfrac{}{X \Rightarrow Y \Rightarrow Z, X \Rightarrow Y, X \vdash X \Rightarrow Y}\,\mathsf{ax} \qquad \dfrac{}{X \Rightarrow Y \Rightarrow Z, X \Rightarrow Y, X \vdash X}\,\mathsf{ax}}{X \Rightarrow Y \Rightarrow Z, X \Rightarrow Y, X \vdash Y}\Rightarrow_e}{X \Rightarrow Y \Rightarrow Z, X \Rightarrow Y, X \vdash Z}\Rightarrow_e}{X \Rightarrow Y \Rightarrow Z, X \Rightarrow Y \vdash X \Rightarrow Z}\Rightarrow_i$$

6. In ND:

$$\dfrac{\dfrac{[Y \Rightarrow Z]^* \qquad \dfrac{(X \Rightarrow X) \Rightarrow Y \qquad \dfrac{[X]^\circ}{X \Rightarrow X}\Rightarrow_i^\circ}{Y}\Rightarrow_e}{Z}\Rightarrow_e}{(Y \Rightarrow Z) \Rightarrow Z}\Rightarrow_i^*$$

In $\mathsf{ND}_{\mathrm{seq}}$:

$$\dfrac{\dfrac{\dfrac{}{Y \Rightarrow Z, (X \Rightarrow X) \Rightarrow Y \vdash Y \Rightarrow Z}\,\mathsf{ax} \qquad \dfrac{\dfrac{}{Y \Rightarrow Z, (X \Rightarrow X) \Rightarrow Y \vdash (X \Rightarrow X) \Rightarrow Y}\,\mathsf{ax} \qquad \dfrac{\dfrac{}{Y \Rightarrow Z, (X \Rightarrow X) \Rightarrow Y, X \vdash X}\,\mathsf{ax}}{Y \Rightarrow Z, (X \Rightarrow X) \Rightarrow Y \vdash X \Rightarrow X}\Rightarrow_i}{Y \Rightarrow Z, (X \Rightarrow X) \Rightarrow Y \vdash Y}\Rightarrow_e}{Y \Rightarrow Z, (X \Rightarrow X) \Rightarrow Y \vdash Z}\Rightarrow_e}{(X \Rightarrow X) \Rightarrow Y \vdash (Y \Rightarrow Z) \Rightarrow Z}\Rightarrow_i$$

Exercise 2

Show that $\nvdash (X \Rightarrow Y) \Rightarrow X$, i.e. $(X \Rightarrow Y) \Rightarrow X$ is not derivable with no hypotheses.

Solution to Exercise 2

Suppose by absurd that $(X \Rightarrow Y) \Rightarrow X$ is derivable in ND with no hypothesis. The last rule of the derivation cannot be either an hypothesis (because there are no hypotheses) or $\Rightarrow_e$ (otherwise it would be it would contradict the subformula property), hence it could only be $\Rightarrow_i$ discharging the hypothesis $X \Rightarrow Y$, that is,

$$\dfrac{\begin{array}{c}[X \Rightarrow Y]^* \\ \vdots \\ X\end{array}}{(X \Rightarrow Y) \Rightarrow X}\Rightarrow_i^*$$

The rule whose conclusion is X cannot be either $\Rightarrow_i$ (otherwise its conclusion should be an arrow) or an hypothesis (because there is no hypothesis X), hence it could only be $\Rightarrow_e$ with premises $A \Rightarrow X$ and A for some formula A, that is,

$$\dfrac{\dfrac{\begin{array}{c}[X \Rightarrow Y]^* \\ \vdots \\ A \Rightarrow X\end{array} \qquad \begin{array}{c}[X \Rightarrow Y]^* \\ \vdots \\ A\end{array}}{X}\Rightarrow_e}{(X \Rightarrow Y) \Rightarrow X}\Rightarrow_i^*$$

For the subformula property applied to the derivation whose conclusion is X, A could only be a subformula of X or $X \Rightarrow Y$, that is,

- either $A = X$, but then $A \Rightarrow X = X \Rightarrow X$ is a formula of that derivation that is not a subformula of X or $X \Rightarrow Y$, which contradicts the subformula property;
- or $A = Y$, but then $A \Rightarrow X = Y \Rightarrow X$ is a formula of that derivation that is not a subformula of X or $X \Rightarrow Y$, which contradicts the subformula property;
- or $A = X \Rightarrow Y$, but then $A \Rightarrow X = (X \Rightarrow Y) \Rightarrow X$ is a formula of that derivation that is not a subformula of X or $X \Rightarrow Y$, which contradicts the subformula property.

Therefore, there is no derivation of $(X \Rightarrow Y) \Rightarrow X$ with no hypotheses.

Exercise 3

Perform all passible cut-elimination steps from the derivation on p. 24 of Day 1 slides, until you get a derivation without redexes. Is it always the same?

Solution to Exercise 3

The derivation on p. 24 of Day 1 slides is $\mathcal{D}$ below, where there are two redexes, marked as blue and red.

$$\dfrac{\dfrac{\dfrac{[(X \Rightarrow X) \Rightarrow (B \Rightarrow X \Rightarrow X)]^\dagger \quad [X \Rightarrow X]^\circ}{B \Rightarrow (X \Rightarrow X)}\Rightarrow_e \qquad [(X \Rightarrow X) \Rightarrow B]^* \quad [X \Rightarrow X]^\circ}{B}\Rightarrow_e}{\dfrac{X \Rightarrow X}{(X \Rightarrow X) \Rightarrow (X \Rightarrow X)}\Rightarrow_i^\circ \qquad \dfrac{[X]^\bullet}{X \Rightarrow X}\Rightarrow_i^\bullet}\Rightarrow_e$$

$$\dfrac{\dfrac{\dfrac{X \Rightarrow X}{((X \Rightarrow X) \Rightarrow B) \Rightarrow (X \Rightarrow X)}\Rightarrow_i^*}{((X \Rightarrow X) \Rightarrow (B \Rightarrow X \Rightarrow X)) \Rightarrow ((X \Rightarrow X) \Rightarrow B) \Rightarrow (X \Rightarrow X))}\Rightarrow_i^\dagger \qquad \dfrac{\dfrac{[X \Rightarrow X]^\dagger}{B \Rightarrow X \Rightarrow X}\Rightarrow_i}{(X \Rightarrow X) \Rightarrow (B \Rightarrow X \Rightarrow X)}\Rightarrow_i^\dagger}{((X \Rightarrow X) \Rightarrow B) \Rightarrow (X \Rightarrow X)}\Rightarrow_e$$

If the red redex in $\mathcal{D}$ is fired, then $\mathcal{D}$ reduces to the derivation $\mathcal{D}_1$ below.

$$\dfrac{\dfrac{\dfrac{[(X \Rightarrow X) \Rightarrow (B \Rightarrow X \Rightarrow X)]^\dagger \quad \dfrac{[X]^\bullet}{X \Rightarrow X}\Rightarrow_i^\bullet}{B \Rightarrow (X \Rightarrow X)}\Rightarrow_e \qquad [(X \Rightarrow X) \Rightarrow B]^* \quad \dfrac{[X]^\bullet}{X \Rightarrow X}\Rightarrow_i^\bullet}{B}\Rightarrow_e}{\dfrac{X \Rightarrow X}{((X \Rightarrow X) \Rightarrow B) \Rightarrow (X \Rightarrow X)}\Rightarrow_i^*}$$

$$\dfrac{\dfrac{\dfrac{X \Rightarrow X}{((X \Rightarrow X) \Rightarrow B) \Rightarrow (X \Rightarrow X)}\Rightarrow_i^*}{((X \Rightarrow X) \Rightarrow (B \Rightarrow X \Rightarrow X)) \Rightarrow ((X \Rightarrow X) \Rightarrow B) \Rightarrow (X \Rightarrow X))}\Rightarrow_i^\dagger \qquad \dfrac{\dfrac{[X \Rightarrow X]^\dagger}{B \Rightarrow X \Rightarrow X}\Rightarrow_i}{(X \Rightarrow X) \Rightarrow (B \Rightarrow X \Rightarrow X)}\Rightarrow_i^\dagger}{((X \Rightarrow X) \Rightarrow B) \Rightarrow (X \Rightarrow X)}\Rightarrow_e$$

If the blue redex in $\mathcal{D}_1$ is fired, then $\mathcal{D}_1$ reduces to the derivation $\mathcal{D}_1'$ below, with a new green redex.

$$\dfrac{\dfrac{\dfrac{\dfrac{[X \Rightarrow X]^{\dagger}}{B \Rightarrow X \Rightarrow X}\Rightarrow_i}{(X \Rightarrow X) \Rightarrow (B \Rightarrow X \Rightarrow X)}\Rightarrow_i{}^{\dagger} \quad \dfrac{[X]^{\bullet}}{X \Rightarrow X}\Rightarrow_i^{\bullet}}{B \Rightarrow (X \Rightarrow X)}\Rightarrow_e \quad \dfrac{[(X \Rightarrow X) \Rightarrow B]^{*} \quad \dfrac{[X]^{\bullet}}{X \Rightarrow X}\Rightarrow_i^{\bullet}}{B}\Rightarrow_e}{X \Rightarrow X}\Rightarrow_i^{*}}{((X \Rightarrow X) \Rightarrow B) \Rightarrow (X \Rightarrow X)}$$

If the green redex in $\mathcal{D}_1'$ is fired, then $\mathcal{D}_1'$ reduces to derivation $\mathcal{D}_1''$ below, with a new gray redex.

$$\dfrac{\dfrac{\dfrac{[X]^{\bullet}}{X \Rightarrow X}\Rightarrow_i^{\bullet}}{B \Rightarrow X \Rightarrow X}\Rightarrow_i \quad \dfrac{[(X \Rightarrow X) \Rightarrow B]^{*} \quad \dfrac{[X]^{\bullet}}{X \Rightarrow X}\Rightarrow_i^{\bullet}}{B}\Rightarrow_e}{X \Rightarrow X}\Rightarrow_i^{*}}{((X \Rightarrow X) \Rightarrow B) \Rightarrow (X \Rightarrow X)}$$

If the gray redex in $\mathcal{D}_1''$ is fired, then $\mathcal{D}_1''$ reduces to derivation $\mathcal{D}_0$ below, which is without redexes.

$$\dfrac{\dfrac{[X]^{\bullet}}{X \Rightarrow X}\Rightarrow_i^{\bullet}}{((X \Rightarrow X) \Rightarrow B) \Rightarrow (X \Rightarrow X)}\Rightarrow_i$$

If the blue redex in $\mathcal{D}$ is fired, then $\mathcal{D}$ reduces to the derivation $\mathcal{D}_2$ below, with a new green redex.

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{[X \Rightarrow X]^{\dagger}}{B \Rightarrow X \Rightarrow X}\Rightarrow_i}{(X \Rightarrow X) \Rightarrow (B \Rightarrow X \Rightarrow X)}\Rightarrow_i{}^{\dagger} \quad [X \Rightarrow X]^{\circ}}{B \Rightarrow (X \Rightarrow X)}\Rightarrow_e \quad \dfrac{[(X \Rightarrow X) \Rightarrow B]^{*} \quad [X \Rightarrow X]^{\circ}}{B}\Rightarrow_e}{X \Rightarrow X}}{(X \Rightarrow X) \Rightarrow (X \Rightarrow X)}\Rightarrow_i{}^{\circ} \quad \dfrac{[X]^{\bullet}}{X \Rightarrow X}\Rightarrow_i^{\bullet}}{X \Rightarrow X}\Rightarrow_e}{((X \Rightarrow X) \Rightarrow B) \Rightarrow (X \Rightarrow X)}\Rightarrow_i^{*}$$

If the red redex in $\mathcal{D}_2$ is fired, then $\mathcal{D}_2$ reduces to the derivation $\mathcal{D}_{21}$ below.

$$\dfrac{\dfrac{\dfrac{\dfrac{[X \Rightarrow X]^{\dagger}}{B \Rightarrow X \Rightarrow X}\Rightarrow_i}{(X \Rightarrow X) \Rightarrow (B \Rightarrow X \Rightarrow X)}\Rightarrow_i{}^{\dagger} \quad \dfrac{[X]^{\bullet}}{X \Rightarrow X}\Rightarrow_i^{\bullet}}{B \Rightarrow (X \Rightarrow X)}\Rightarrow_e \quad \dfrac{[(X \Rightarrow X) \Rightarrow B]^{*} \quad \dfrac{[X]^{\bullet}}{X \Rightarrow X}\Rightarrow_i^{\bullet}}{B}\Rightarrow_e}{X \Rightarrow X}\Rightarrow_i^{*}}{((X \Rightarrow X) \Rightarrow B) \Rightarrow (X \Rightarrow X)}$$

If the green redex in $\mathcal{D}_{21}$ is fired, then $\mathcal{D}_{21}$ reduces to the derivation $\mathcal{D}_1''$ already shown above.
If the green redex in $\mathcal{D}_2$ is fired, then $\mathcal{D}_2$ reduces to the derivation $\mathcal{D}_{22}$ below, with a new gray redex.

$$\dfrac{\dfrac{\dfrac{\dfrac{[X \Rightarrow X]^{\circ}}{B \Rightarrow X \Rightarrow X}\Rightarrow_i \quad \dfrac{[(X \Rightarrow X) \Rightarrow B]^{*} \quad [X \Rightarrow X]^{\circ}}{B}\Rightarrow_e}{X \Rightarrow X}}{(X \Rightarrow X) \Rightarrow (X \Rightarrow X)}\Rightarrow_i{}^{\circ} \quad \dfrac{[X]^{\bullet}}{X \Rightarrow X}\Rightarrow_i^{\bullet}}{X \Rightarrow X}\Rightarrow_e}{((X \Rightarrow X) \Rightarrow B) \Rightarrow (X \Rightarrow X)}\Rightarrow_i^{*}$$

If the red redex in $\mathcal{D}_{22}$ is fired, then $\mathcal{D}_{22}$ reduces to the derivation $\mathcal{D}_{221}$ below.

$$\dfrac{\dfrac{\dfrac{[X]^{\bullet}}{X \Rightarrow X}\Rightarrow_i^{\bullet}}{B \Rightarrow X \Rightarrow X}\Rightarrow_i \quad \dfrac{[(X \Rightarrow X) \Rightarrow B]^{*} \quad \dfrac{[X]^{\bullet}}{X \Rightarrow X}\Rightarrow_i^{\bullet}}{B}\Rightarrow_e}{X \Rightarrow X}\Rightarrow_i^{*}}{((X \Rightarrow X) \Rightarrow B) \Rightarrow (X \Rightarrow X)}$$

If the gray redex in $\mathcal{D}_{221}$ is fired, then $\mathcal{D}_{221}$ reduces to the derivation $\mathcal{D}_0$ below, which is without redexes.

$$\dfrac{\dfrac{[X]^\bullet}{X \Rightarrow X}\Rightarrow_i^\bullet}{((X \Rightarrow X) \Rightarrow B) \Rightarrow (X \Rightarrow X)}\Rightarrow_i$$

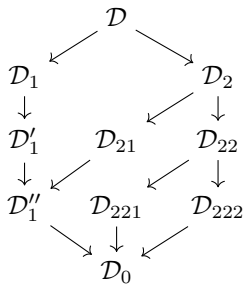
If the gray redex in $\mathcal{D}_{22}$ is fired, then $\mathcal{D}_{22}$ reduces to the derivation $\mathcal{D}_{222}$ below.

$$\dfrac{\dfrac{\dfrac{[X \Rightarrow X]^\circ}{(X \Rightarrow X) \Rightarrow (X \Rightarrow X)}\Rightarrow_i^\circ \quad \dfrac{[X]^\bullet}{X \Rightarrow X}\Rightarrow_i^\bullet}{X \Rightarrow X}\Rightarrow_e}{((X \Rightarrow X) \Rightarrow B) \Rightarrow (X \Rightarrow X)}\Rightarrow_i^*$$

If the red redex in $\mathcal{D}_{222}$ is fired, then $\mathcal{D}_{222}$ reduces to the derivation $\mathcal{D}_0$ below, which is without redexes.

$$\dfrac{\dfrac{[X]^\bullet}{X \Rightarrow X}\Rightarrow_i^\bullet}{((X \Rightarrow X) \Rightarrow B) \Rightarrow (X \Rightarrow X)}\Rightarrow_i$$

All possible cut-elimination steps from $\mathcal{D}$ are the following:



In any case, every reduction sequence eventually reaches the same derivation $\mathcal{D}_0$ with no redexes.

Exercise 4

Order the following multisets over $\mathbb{N}$ according to the (strict) multiset order $\prec_{\text{mul}}$.

$$[1,1] \quad [0,2] \quad [1] \quad [0,0,2] \quad [\,] \quad [0,3] \quad [0,2,2]$$

Solution to Exercise 4

$$[\,] \prec_{\text{mul}} [1] \prec_{\text{mul}} [1,1] \prec_{\text{mul}} [0,2] \prec_{\text{mul}} [0,0,2] \prec_{\text{mul}} [0,2,2] \prec_{\text{mul}} [0,3].$$

Exercise 5

Prove in a rigorous way the proposition on p. 15 of Day 1 slides.

Solution to Exercise 5

Proposition. *Let Γ be a finite multiset of formulas and A be a formula: $\Gamma \vdash A$ in ND if and only if the sequent $\Gamma \vdash A$ is derivable in ND_{seq}.*

Proof. $\Rightarrow$**:** By induction on the number of rules of the smallest derivation $\mathcal{D}$ in ND proving that $\Gamma \vdash A$. Cases:

- $\mathcal{D}$ is just an hypothesis, that is, $\mathcal{D} = A$ and so $\Gamma = \Gamma', A$ for any finite multiset Γ'. Then, the derivation $\mathcal{D}_{\text{seq}}$ below derives the sequent $\Gamma \vdash A$ in ND_{seq}.

$$\mathcal{D}_{\text{seq}} \quad = \quad \dfrac{}{\Gamma', A \vdash A}\,\mathsf{ax}$$

- The last rule in $\mathcal{D}$ is $\Rightarrow_i$, that is, $A = B \Rightarrow C$ and

$$\mathcal{D} \quad = \quad \dfrac{\begin{array}{c}[B]^* \\ \vdots\,\mathcal{D}' \\ C\end{array}}{B \Rightarrow C}\Rightarrow_i^*$$

where $\mathcal{D}'$ is the smallest derivation in ND that proves that $\Gamma, B \vdash C$, by minimality of $\mathcal{D}$. By induction hypothesis applied to $\mathcal{D}'$, there is a derivation $\mathcal{D}'_{\text{seq}}$ in ND_{seq} of the sequent $\Gamma, B \vdash C$. Then, the derivation $\mathcal{D}_{\text{seq}}$ below derives the sequent $\Gamma \vdash A$ in ND_{seq}.

$$\mathcal{D}_{\text{seq}} \quad = \quad \dfrac{\begin{array}{c}\vdots\,\mathcal{D}'_{\text{seq}}\\ \Gamma, B \vdash C\end{array}}{\Gamma \vdash B \Rightarrow C}\Rightarrow_i$$

- The last rule in $\mathcal{D}$ is $\Rightarrow_e$, that is, for some formula B

$$\mathcal{D} \quad = \quad \dfrac{\begin{array}{cc}\vdots\,\mathcal{D}' & \vdots\,\mathcal{D}''\\ B \Rightarrow A & B\end{array}}{A}\Rightarrow_e$$

where $\mathcal{D}'$ and $\mathcal{D}''$ are the smallest derivation in ND that prove that $\Gamma \vdash B \Rightarrow A$ and $\Gamma \vdash B$, respectively, by minimality of $\mathcal{D}$. By induction hypothesis applied to $\mathcal{D}'$ and $\mathcal{D}''$, respectively, there are derivations $\mathcal{D}'_{\text{seq}}$ and $\mathcal{D}''_{\text{seq}}$ in ND_{seq} of the sequents $\Gamma \vdash B \Rightarrow A$ and $\Gamma \vdash B$. Then, the derivation $\mathcal{D}_{\text{seq}}$ below derives the sequent $\Gamma \vdash A$ in ND_{seq}.

$$\mathcal{D}_{\text{seq}} \quad = \quad \dfrac{\begin{array}{cc}\vdots\,\mathcal{D}'_{\text{seq}} & \vdots\,\mathcal{D}''_{\text{seq}}\\ \Gamma \vdash B \Rightarrow A & \Gamma \vdash B\end{array}}{\Gamma \vdash A}\Rightarrow_e$$

$\Leftarrow$**:** By induction on the number of rules of the smallest derivation $\mathcal{D}$ in ND_{seq} proving the sequent $\Gamma \vdash A$. Cases:

- The last rule of $\mathcal{D}$ is ax, that is,

$$\mathcal{D} \quad = \quad \dfrac{}{\Gamma', A \vdash A}\,\mathsf{ax}$$

where $\Gamma = \Gamma', A$ for some finite multiset Γ'. Then, the derivation $\mathcal{D}_0 = A$ proves that $\Gamma \vdash A$ in ND.

- The last rule in $\mathcal{D}$ is $\Rightarrow_i$, that is, $A = B \Rightarrow C$ and

$$\mathcal{D} \quad = \quad \dfrac{\begin{array}{c}\vdots\,\mathcal{D}'\\ \Gamma, B \vdash C\end{array}}{\Gamma \vdash B \Rightarrow C}\Rightarrow_i$$

where $\mathcal{D}'$ is the smallest derivation in ND_{seq} of the sequent $\Gamma, B \vdash C$, by minimality of $\mathcal{D}$. By induction hypothesis applied to $\mathcal{D}'$, there is a derivation $\mathcal{D}'_0$ in ND that proves that $\Gamma, B \vdash C$. Then, the derivation $\mathcal{D}_0$ below proves that $\Gamma \vdash A$ in ND.

$$\mathcal{D} \quad = \quad \dfrac{\begin{array}{c}[B]^*\\ \vdots\,\mathcal{D}'\\ C\end{array}}{B \Rightarrow C}\Rightarrow^*_i$$

- The last rule in $\mathcal{D}$ is $\Rightarrow_e$, that is, for some formula B

$$\mathcal{D}_{\text{seq}} \quad = \quad \dfrac{\begin{array}{cc}\vdots\,\mathcal{D}' & \vdots\,\mathcal{D}''\\ \Gamma \vdash B \Rightarrow A & \Gamma \vdash B\end{array}}{\Gamma \vdash A}\Rightarrow_e$$

where $\mathcal{D}'$ and $\mathcal{D}''$ are the smallest derivation in ND_{seq} that prove the sequents $\Gamma \vdash B \Rightarrow A$ and $\Gamma \vdash B$, respectively, by minimality of $\mathcal{D}$. By induction hypothesis applied to $\mathcal{D}'$ and $\mathcal{D}''$, respectively, there are derivations $\mathcal{D}'_0$ and $\mathcal{D}''_0$ in ND that prove $\Gamma \vdash B \Rightarrow A$ and $\Gamma \vdash B$. Then, the derivation $\mathcal{D}_0$ below prove that $\Gamma \vdash A$ in ND.

$$\mathcal{D} \quad = \quad \dfrac{\begin{array}{cc}\vdots\,\mathcal{D}' & \vdots\,\mathcal{D}''\\ B \Rightarrow A & B\end{array}}{A}\Rightarrow_e$$

□

Exercise 6

For any formula B, prove that if $\Gamma \vdash A$ is derivable in ND_{seq}, then so is $\Gamma, B \vdash A$.

Solution to Exercise 6

By induction on the number of rules of the smallest derivation $\mathcal{D}$ in $\mathsf{ND}_{\mathrm{seq}}$ proving the sequent $\Gamma \vdash A$. Cases:

- The last rule of $\mathcal{D}$ is ax, that is,

$$\mathcal{D} \quad = \quad \dfrac{}{\Gamma', A \vdash A}\,\mathsf{ax}$$

where $\Gamma = \Gamma', A$ for some finite multiset Γ'. Then, the derivation below proves the sequent $\Gamma, B \vdash A$ in $\mathsf{ND}_{\mathrm{seq}}$.

$$\dfrac{}{\Gamma', B, A \vdash A}\,\mathsf{ax}$$

- The last rule in $\mathcal{D}$ is $\Rightarrow_i$, that is, $A = D \Rightarrow C$ and

$$\mathcal{D} \quad = \quad \dfrac{\begin{array}{c}\vdots\,\mathcal{D}'\\ \Gamma, D \vdash C\end{array}}{\Gamma \vdash D \Rightarrow C}\Rightarrow_i$$

where $\mathcal{D}'$ is the smallest derivation in $\mathsf{ND}_{\mathrm{seq}}$ of the sequent $\Gamma, D \vdash C$, by minimality of $\mathcal{D}$. By induction hypothesis applied to $\mathcal{D}'$, there is a derivation $\mathcal{D}_0$ in $\mathsf{ND}_{\mathrm{seq}}$ that proves the sequent $\Gamma, B, D \vdash C$. Then, the derivation below proves the sequent $\Gamma, B \vdash A$ in $\mathsf{ND}_{\mathrm{seq}}$.

$$\dfrac{\begin{array}{c}\vdots\,\mathcal{D}_0\\ \Gamma, B, D \vdash C\end{array}}{\Gamma, B \vdash D \Rightarrow C}\Rightarrow_i$$

- The last rule in $\mathcal{D}$ is $\Rightarrow_e$, that is, for some formula C

$$\mathcal{D}_{\mathrm{seq}} \quad = \quad \dfrac{\begin{array}{cc}\vdots\,\mathcal{D}' & \vdots\,\mathcal{D}''\\ \Gamma \vdash C \Rightarrow A & \Gamma \vdash C\end{array}}{\Gamma \vdash A}\Rightarrow_e$$

where $\mathcal{D}'$ and $\mathcal{D}''$ are the smallest derivations in $\mathsf{ND}_{\mathrm{seq}}$ that prove the sequents $\Gamma \vdash C \Rightarrow A$ and $\Gamma \vdash C$, respectively, by minimality of $\mathcal{D}$. By induction hypothesis applied to $\mathcal{D}'$ and $\mathcal{D}''$, respectively, there are derivations $\mathcal{D}_1$ and $\mathcal{D}_2$ in $\mathsf{ND}_{\mathrm{seq}}$ that prove the sequents $\Gamma, B \vdash C \Rightarrow A$ and $\Gamma \vdash C$. Then, the derivation below prove the sequent $\Gamma, B \vdash A$ in $\mathsf{ND}_{\mathrm{seq}}$.

$$\dfrac{\begin{array}{cc}\vdots\,\mathcal{D}_1 & \vdots\,\mathcal{D}_2\\ \Gamma, B \vdash C \Rightarrow A & \Gamma, B \vdash C\end{array}}{\Gamma, B \vdash A}\Rightarrow_e$$

Exercise 7

For any formula B, prove that if $\Gamma, B, B \vdash A$ is derivable in $\mathsf{ND}_{\mathrm{seq}}$ then so is $\Gamma, B \vdash A$.

Solution to Exercise 7

By induction on the number of rules of the smallest derivation $\mathcal{D}$ in $\mathsf{ND}_{\mathrm{seq}}$ proving the sequent $\Gamma, B, B \vdash A$. Cases:

- The last rule of $\mathcal{D}$ is ax, that is,

$$\mathcal{D} \quad = \quad \dfrac{}{\Gamma', B, B, A \vdash A}\,\mathsf{ax}$$

where $\Gamma = \Gamma', A$ for some finite multiset Γ'. Then, the derivation below proves the sequent $\Gamma, B \vdash A$ in $\mathsf{ND}_{\mathrm{seq}}$.

$$\dfrac{}{\Gamma', B, A \vdash A}\,\mathsf{ax}$$

- The last rule in $\mathcal{D}$ is $\Rightarrow_i$, that is, $A = D \Rightarrow C$ and

$$\mathcal{D} \quad = \quad \dfrac{\begin{array}{c}\vdots\,\mathcal{D}'\\ \Gamma, B, D \vdash C\end{array}}{\Gamma, B, B \vdash D \Rightarrow C}\Rightarrow_i$$

where $\mathcal{D}'$ is the smallest derivation in $\mathsf{ND}_{\mathrm{seq}}$ of the sequent $\Gamma, B, B, D \vdash C$, by minimality of $\mathcal{D}$. By induction hypothesis applied to $\mathcal{D}'$, there is a derivation $\mathcal{D}_0$ in $\mathsf{ND}_{\mathrm{seq}}$ that proves the sequent $\Gamma, B, D \vdash C$. Then, the derivation below proves the sequent $\Gamma, B \vdash A$ in $\mathsf{ND}_{\mathrm{seq}}$.

$$\dfrac{\begin{array}{c}\vdots\,\mathcal{D}_0\\ \Gamma, B, D \vdash C\end{array}}{\Gamma, B \vdash D \Rightarrow C}\Rightarrow_i$$

- The last rule in $\mathcal{D}$ is $\Rightarrow_e$, that is, for some formula C

$$\mathcal{D}_{\mathrm{seq}} \quad = \quad \dfrac{\begin{array}{cc}\vdots\,\mathcal{D}' & \vdots\,\mathcal{D}''\\ \Gamma, B, B \vdash C \Rightarrow A & \Gamma, B, B \vdash C\end{array}}{\Gamma, B, B \vdash A}\Rightarrow_e$$

where $\mathcal{D}'$ and $\mathcal{D}''$ are the smallest derivations in $\mathsf{ND}_{\mathrm{seq}}$ that prove the sequents $\Gamma, B, B \vdash C \Rightarrow A$ and $\Gamma, B, B \vdash C$, respectively, by minimality of $\mathcal{D}$. By induction hypothesis applied to $\mathcal{D}'$ and $\mathcal{D}''$, respectively, there are derivations $\mathcal{D}_1$ and $\mathcal{D}_2$ in $\mathsf{ND}_{\mathrm{seq}}$ that prove the sequents $\Gamma, B \vdash C \Rightarrow A$ and $\Gamma \vdash C$. Then, the derivation below proves the sequent $\Gamma, B \vdash A$ in $\mathsf{ND}_{\mathrm{seq}}$.

$$\dfrac{\begin{array}{cc}\vdots\,\mathcal{D}_1 & \vdots\,\mathcal{D}_2\\ \Gamma, B \vdash C \Rightarrow A & \Gamma, B \vdash C\end{array}}{\Gamma, B \vdash A}\Rightarrow_e$$